Generate Invoice and Record Payment – System Operation Contracts

Contract CO4 – Search Completed Appointments

Field	Answer
Operation:	searchCompletedAppointments(searchCriteria)
Use Case:	Generate Invoice and Record Payment
Preconditions:	Administrative Staff is authenticated and authorized to search appointments. searchCriteria contains at least one of: insurance number, date of birth, phone number, or partial name
Success Postconditions:	System returns the completed appointments, matching the supplied search criteria, that have not yet been invoiced. No domain objects are modified
Failure Postconditions:	No matching appointments are returned. No domain objects are modified

Contract CO5 – Select Appointment

Field	Answer
Operation:	selectAppointment(appointmentId)
Use Case:	Generate Invoice and Record Payment
Preconditions:	Administrative Staff is authenticated and authorized to manage invoices. A validly formatted appointment identifier has been provided
Success Postconditions:	System confirms the Appointment is completed, is an uninsured service (patient not covered by RAMQ for the service), and has no associated Invoice, and returns the appointment and patient billing information. No domain objects are modified
Failure Postconditions:	The appointment is rejected as not billable (covered by RAMQ, not completed, or already invoiced). No domain objects are modified. The system returns a failure indication

Contract CO6 – Generate Invoice

Field	Answer
Operation:	generateInvoice(appointmentId, applicableFees)
Use Case:	Generate Invoice and Record Payment
Preconditions:	Administrative Staff is authenticated and authorized to manage invoices. The appointment has been selected and verified as billable (CO5). applicableFees contains at least one fee with a valid, non-negative amount
Success Postconditions:	A new Invoice instance is created (instance creation) The Invoice's applicable_fees is set to the supplied fees and its status is set to Unpaid (attribute modification)

	The Invoice is associated with the Appointment and the Patient (association formed) A new ActivityRecord instance is created to document the invoice generation and associated with the responsible staff member (instance creation, association formed)
Failure Postconditions:	No Invoice instance is created. No ActivityRecord is created. The system returns a failure indication

Contract CO7 – Record Payment

Field	Answer
Operation:	recordPayment(invoiceId, amount, paymentMethod)
Use Case:	Generate Invoice and Record Payment
Preconditions:	Administrative Staff is authenticated and authorized to register payments. An Invoice with status Unpaid exists for the provided invoice identifier. amount and paymentMethod have been provided
Success Postconditions:	A new Payment instance is created with the supplied amount, payment method and the current date and time (instance creation) The Payment is associated with the Invoice (association formed) The Invoice's status is changed to Paid (attribute modification) A new ActivityRecord instance is created to document the payment and associated with the responsible staff member (instance creation, association formed)
Failure Postconditions:	No Payment instance is created. The Invoice's status remains Unpaid. No ActivityRecord is created. The system returns a failure indication with the remaining balance